

UECS2094/ UECS2194 WEB APPLICATION DEVELOPMENT

PRACTICAL ASSESSMENT

Feb 2025 Trimester

Course Outcomes.

In this assessment, you are assessed based on the following course outcomes:

CO5	Demonstrate the skill to use dependency managers to manage web application dependencies in web application development.
-----	---

Assessment Contribution

This assessment contributes 15% to the overall assessment for this course.

Instructions:

1. This is an open-book test. Please ensure that you answer all questions. You have 3 hours to complete the test.
2. Use the provided MS Word file as your answer sheet. Fill in your answers and save the file as a PDF. Name the file using the following format: Programme_StudentID_Name.pdf (e.g., SE_1234567_JohnTan.pdf).
3. Upload the saved PDF file to WBLE for submission before the end of the test.
4. To avoid losing any work, save your answers and project files frequently to a backup location, such as a pen drive or Google Drive. This is especially important to avoid data loss in case of technical issues.
5. For every webpage, form, database, and files in your project folder, take clear screenshots and attach them to the corresponding section of your answer sheet. For code snippets, you can either paste the code directly into your answer sheet or take a screenshot of the code and attach it to the relevant section.
6. Ensure that all screenshots are clear and fully capture all required elements for each question. Blurry or cropped images may result in lost marks, so please make sure the screenshots are legible.
7. For all questions, ensure that you apply the CSS styling following the Figures provided. The styling should be correctly implemented in your project to meet the requirements of the question.
8. It is your responsibility to verify that all required codes and screenshots are inserted before submitting your answer sheet. Incomplete submissions may result in a deduction of marks.

You are tasked to develop a web-based Study Plan Management system for the students of University Tunku Abdul Rahman (UTAR). The application will allow students to register an account, select courses, and save the selected courses to their profile.

You are required to develop the following features using **HTML**, **CSS**, **JavaScript**, **PHP**, and **MySQL**:

Task 1. Homepage (index.php) – 9 marks

The homepage of the web application should contain:

- a) A welcome message for the student "Welcome to the UTAR Study Plan Management System". (1 mark)
- b) Contain navigation links to:
 - Homepage.
 - Account Registration page.
 - List of Courses page. (6 marks)
- c) A paragraph to provide basic information about the Study Plan Management system. (1 mark)
- d) An image showing the logo of UTAR (promotional-image.jpg). (1 mark)

Note:

- Figure 1 shows the layout of the Homepage.
- The paragraph text and the image promotional-image.jpg are provided in the test question folder.

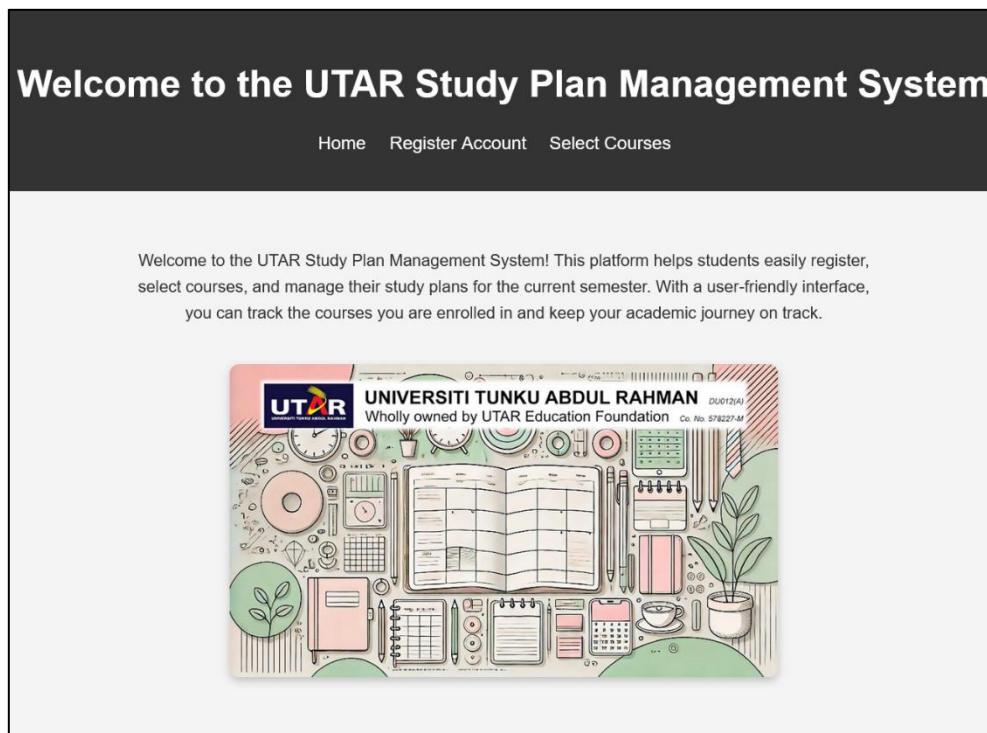


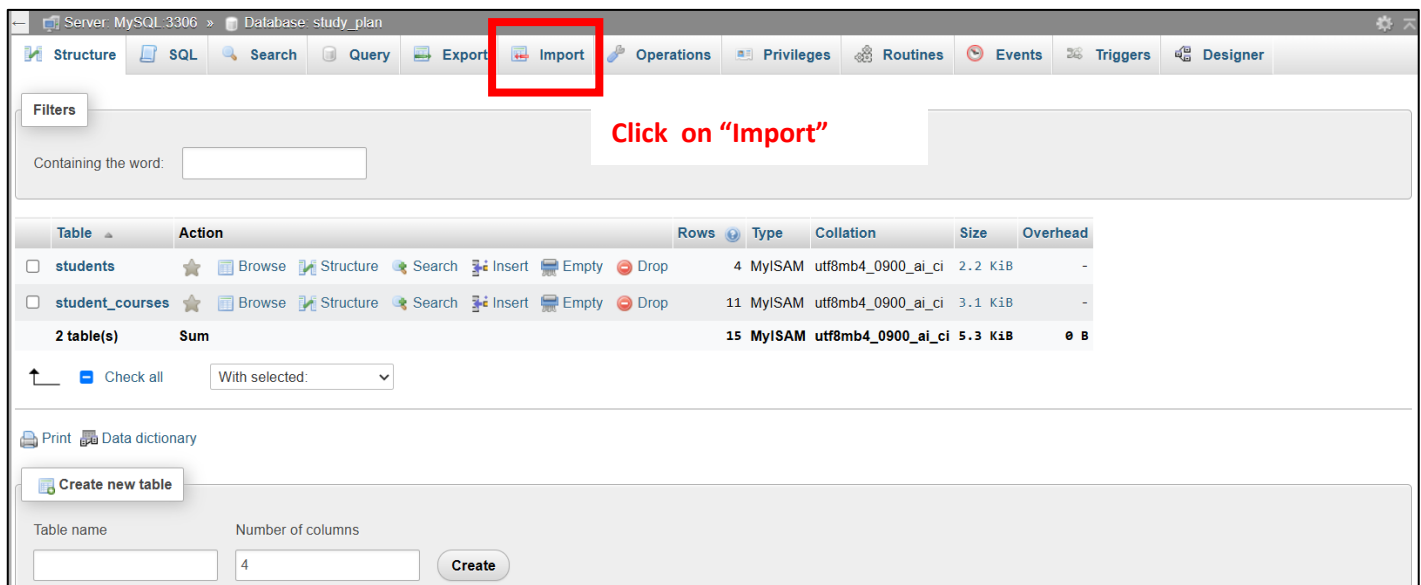
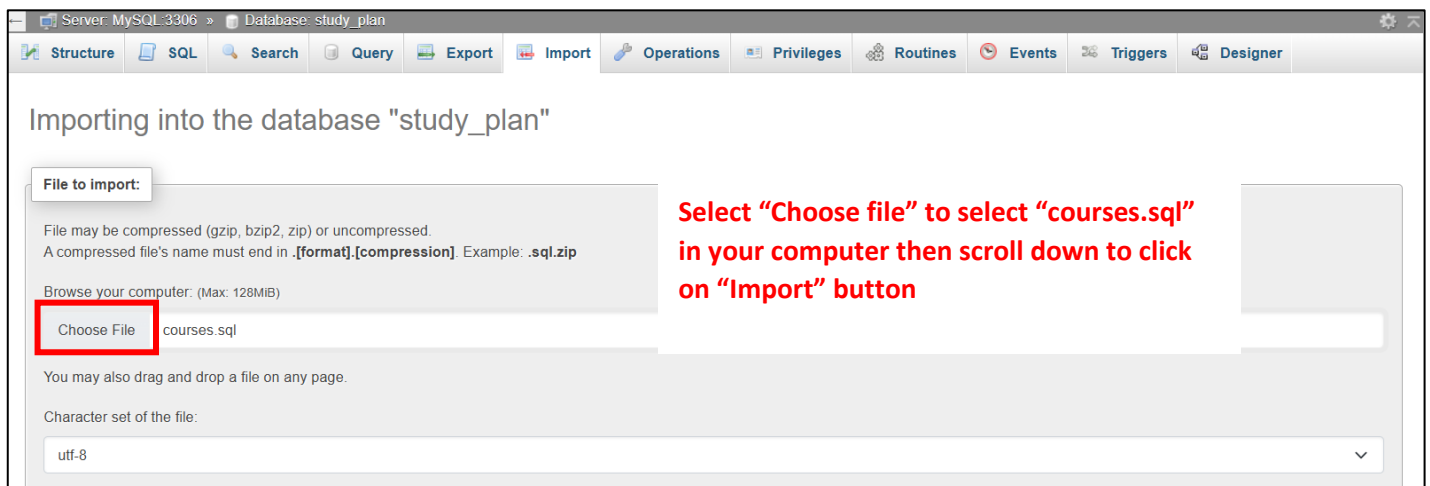
Figure 1: Homepage (index.php)

Task 2. Database (db.sql) – 13 marks

- a) Create a MySQL database named “study_plan” with the collation set to “utf8_general_ci”. (2 marks)
- b) In the study_plan database, create a students table that will store the following information:
- student_id: Integer (Primary Key)
 - name: String (Up to 100 characters)
 - email: String (Up to 100 characters)
 - password: String (Up to 255 characters)
- (4 marks)
- c) Now, create a student_courses table to store the relationship between students and courses, where a student can select many courses and each course can have many students. The student_courses table should have the following structure:
- student_id: Integer (Foreign Key referencing students.student_id)
 - course_id: Integer (Foreign Key referencing a courses table, which is not required to be created but should exist)
 - The combination of student_id and course_id should be the primary key for this table.
- (7 marks)

Note:

- Write the SQL statements and save as “db.sql”. Creating database and table using the “phpMyAdmin” interface is not acceptable.
- After creating the database “study_plan”, import “courses.sql” provided in the question folder to have a table named “courses” imported into the database (refer to Figure 2 and 3).

Figure 2: Import “courses.sql” to the `study_plan` database (Step 1)Figure 3: Import “courses.sql” to the `study_plan` database (Step 2)

Task 3. Account Registration Page (account.php & validation.js) – 18 marks

- a) This page will allow students to create an account by entering the following information:
- **Full Name** (Text input) (1 mark)
 - **Email Address** (Text input) (1 mark)
 - **Student ID** (Numeric input) (1 mark)
 - **Password** (Password input) (1 mark)
- b) The form must have **client-side validation** (validation.js) using JavaScript to ensure:
- All fields are filled out.
 - Email is in a valid format.
 - Password is at least 6 characters long.
 - The Student ID is numeric (with 7 digits length).
 - If there is any error found in the form, an error message is displayed under the respective input field.
- (10 marks)
- c) Upon successful registration, the system should:
- Insert the data into the “students” table in the database “study_plan”.
 - Redirect the user to the **List of Courses page (courses.php)**.
- (4 marks)

Note:

- Use external JavaScript file – validation.js
- Refer to Figure 4 and 5 to observe the layout of the Account Registration page.

The screenshot shows a web application interface for the 'UTAR Study Plan Management System'. At the top, there is a dark header with the title 'Welcome to the UTAR Study Plan Management System' and navigation links: 'Home', 'Register Account', and 'Select Courses'. Below the header, the main content area is titled 'Create an Account'. It contains a registration form with the following fields: 'Full Name:', 'Email Address:', 'Student ID:', and 'Password:'. Each field has a corresponding input box. At the bottom of the form is a green 'Register' button.

Figure 4: Account Registration Page (account.php)

Welcome to the UTAR Study Plan Management System

[Home](#) [Register Account](#) [Select Courses](#)

Create an Account

Full Name:

Full Name is required.

Email Address:

Please enter a valid email address.

Student ID:

Student ID must be exactly 7 digits.

Password:

Password must be at least 6 characters long.

Figure 5: Account Registration Page with error messages displayed

Task 4. List of Courses page (courses.php) – 20 marks

- a) This page will display a list of courses available for the current semester, with checkboxes for selection. The courses should be retrieved from the “courses” table in the “study_plan” database. Each course should have:
- **Course ID**
 - **Course Name**
 - **Course Description**
- (10 marks)
- b) Students can select multiple courses using checkbox from the list and add them to their profile using their **student ID**. The selected courses should be stored in the “student_courses” table created in Task 2 and associated with the student’s profile created in the Account Registration page. (6 marks)
- c) If the student ID does not exist in the database, an error message with a **hyperlink** to the Account Registration page will be displayed (as shown in Figure 7). (2 marks)
- d) After saving the selected course to the “student_courses” table, redirect the student to the Profile page (profile.php), passing the student ID as a query parameter. (2 marks)

Note: You will need to:

- Retrieve course data from the database.
- Allow students to add courses to their profile.
- Ensure that the courses selected are saved into the database under the student’s profile.
- Refer to Figure 6 and 7 to observe the layout of the page

Welcome to the UTAR Study Plan Management System

[Home](#) [Register Account](#) [Select Courses](#)

Select Courses

Student ID:

Available Courses

☐

UECS2194 - Web Application Development

Introduction to basic web development concepts

☐

UECS1104 - Programming and Problem Solving

Introduction to the basic structure of a computer system, types of programming languages, the programming process, and problem-solving concepts.

☐

UECS2333 - Human-Computer Interaction Design

The study of how human, as a user, interact with the hardware and software components of a computer.

☐

UECM1024 - Calculus I

Introduces basic calculus concepts and methods which are of fundamental importance to the study of continuous mathematical models in probability theory, statistics and engineering models.

☐

UECM1314 - Fundamentals of Linear Algebra

Introduction to the most basic concepts in linear algebra. The topics to be discussed in this unit are vectors, matrices, determinants, eigenvalues and eigenvectors, vector spaces, linear transformations and orthogonality.

Save Selected Courses

Figure 6: List of Courses page (courses.php)

Welcome to the UTAR Study Plan Management System

[Home](#) [Register Account](#) [Select Courses](#)

Select Courses

Error: Student ID does not exist. Please [register](#).

Student ID:

Available Courses

☐

UECS2194 - Web Application Development
Introduction to basic web development concepts

☐

UECS1104 - Programming and Problem Solving
Introduction to the basic structure of a computer system, types of programming languages, the programming process, and problem-solving concepts.

☐

UECS2333 - Human-Computer Interaction Design
The study of how human, as a user, interact with the hardware and software components of a computer.

☐

UECM1024 - Calculus I
Introduces basic calculus concepts and methods which are of fundamental importance to the study of continuous mathematical models in probability theory, statistics and engineering models.

☐

UECM1314 - Fundamentals of Linear Algebra
Introduction to the most basic concepts in linear algebra. The topics to be discussed in this unit are vectors, matrices, determinants, eigenvalues and eigenvectors, vector spaces, linear transformations and orthogonality.

Save Selected Courses

Figure 7: List of Courses page (courses.php) – with error message displayed

Task 5. Profile Page (profile.php) – 10 marks

This page will display the student's profile information **after they have added subject from the List of Courses page** (use the student ID as the query parameter). The profile page will include:

- **Student Name**
- **Student Email Address**
- **Student ID**
- A table listing the **courses** the student has selected. The table should display:
 - **Course code**
 - **Course Name**
 - **Course Description**
- If no course is selected by the student, display “No course selected” instead of the table.

(10 marks)

Note:

- The student's profile page should **retrieve their profile data from the database** which matches with the student ID which is passed from the List of Courses page as query parameter in the URL.
- Refer to Figure 8 and 9 to observe the layout of the Profile page.

The screenshot shows a web application interface for the UTAR Study Plan Management System. At the top, a dark header contains the title 'Welcome to the UTAR Study Plan Management System' and navigation links: 'Home', 'Register Account', and 'Select Courses'. Below the header, the main content area is titled 'My Profile'. It displays the user's details: Name: Jason Teoh, Email: jason@1utar.my, and Student ID: 1234567. Under the 'Selected Courses' section, there is a table with three columns: Course Code, Course Name, and Course Description. The table lists three selected courses: UECS2194 (Web Application Development), UECS1104 (Programming and Problem Solving), and UECM1024 (Calculus I).

Course Code	Course Name	Course Description
UECS2194	Web Application Development	Introduction to basic web development concepts
UECS1104	Programming and Problem Solving	Introduction to the basic structure of a computer system, types of programming languages, the programming process, and problem-solving concepts.
UECM1024	Calculus I	Introduces basic calculus concepts and methods which are of fundamental importance to the study of continuous mathematical models in probability theory, statistics and engineering models.

Figure 8: Profile page (profile.php)

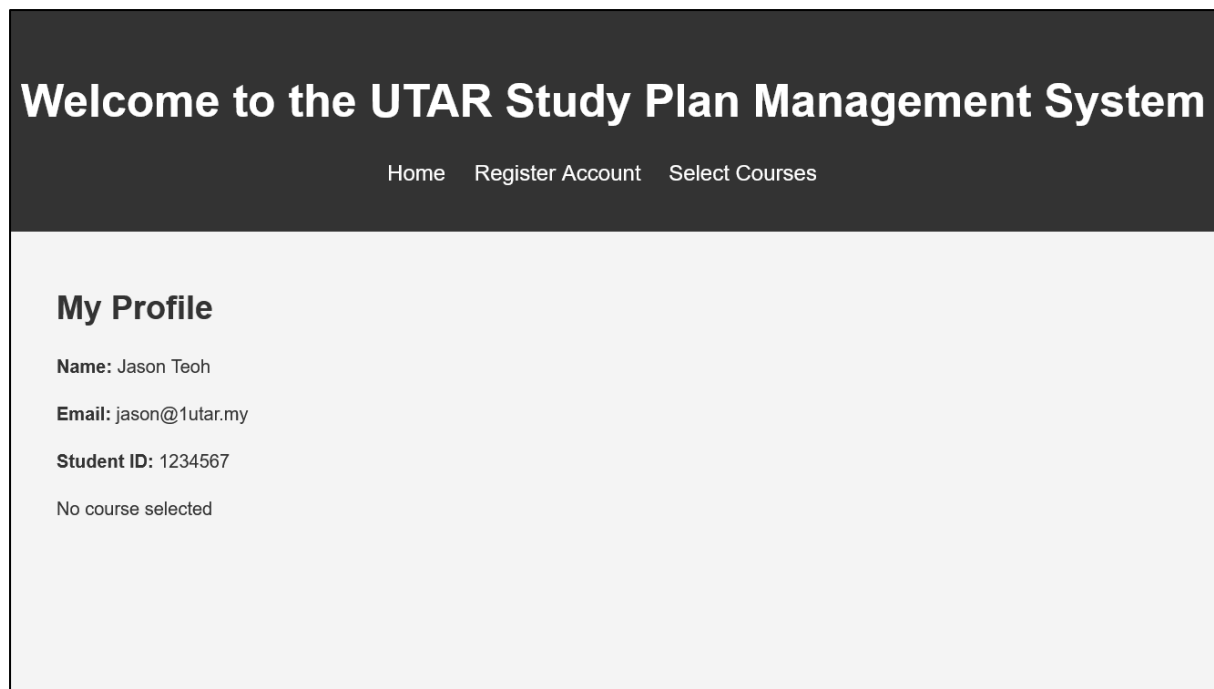


Figure 9: Profile page when no course is selected by the student.

Task 6. Styling the website (styles.css) – 20 marks

Create a CSS stylesheet (styles.css) that will do the following:

- a) Make the style of the Homepage the same as Figure 1.
- b) Make the style of the Account Registration page the same as Figure 4 and 5.
- c) Make the style of the List of Course page the same as Figure 6 and 7.
- d) Make the style of the Profile page the same as Figure 8 and 9.

(20 marks)

Note: Use an external stylesheet (styles.css) and link to the pages in the Study Plan Management web application.

Deliverables:

1. **index.php** - Homepage with navigation.
2. **account.php** - Registration page with client-side validation.
3. **courses.php** - List of courses with the ability to select courses.
4. **profile.php** - Student profile page displaying student details and selected courses.
5. **db.sql** - SQL script to create the database and tables, including relationships.
6. **validation.js** - client-side validation JavaScript code for account.php
7. **styles.css** – Stylesheet for all pages

Bonus question – 5 marks

Enhance the “profile.php” page to include the following functionality:

1. Student ID Check
 - If the student ID is not provided as a query parameter in the URL, display a form asking the user to input their student ID (as shown in Figure 10).
2. Valid Student ID
 - If the student ID exists in the database:
 - Retrieve and display the corresponding profile information of the student.
 - Include a table that lists the courses the student has selected, as shown in Figure 8.
3. Invalid Student ID
 - If the student ID does not exist in the database:
 - Display an error message indicating that the student ID is invalid.
 - Provide a hyperlink to the Account Registration page (as shown in Figure 11) so that the student can register.

Welcome to the UTAR Study Plan Management System

Home Register Account Select Courses

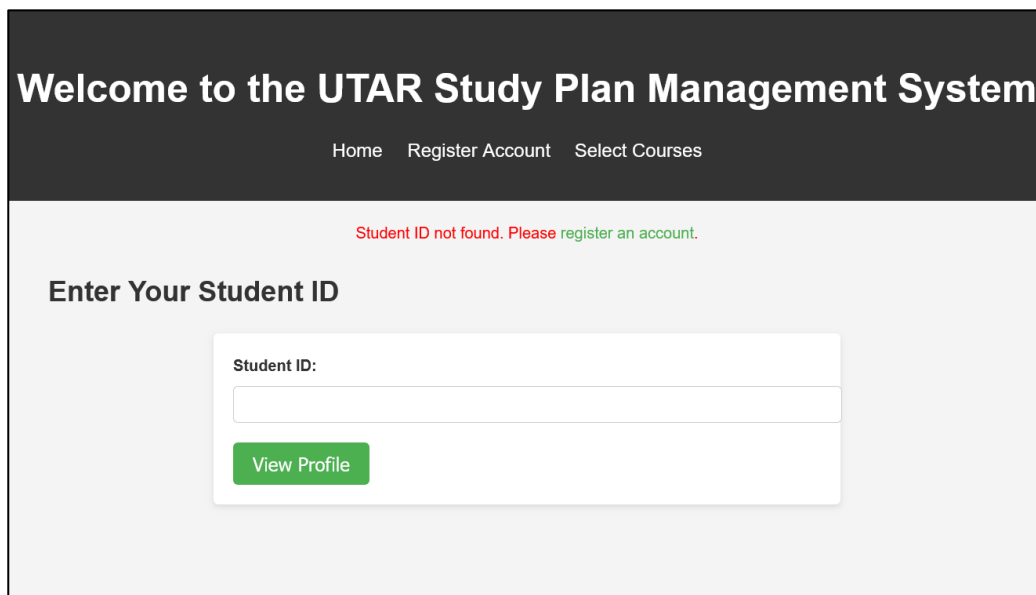
Please enter your Student ID to view your profile.

Enter Your Student ID

Student ID:

View Profile

Figure 10: Profile page when the student ID is not provided as a query parameter.



Welcome to the UTAR Study Plan Management System

[Home](#) [Register Account](#) [Select Courses](#)

Student ID not found. Please [register an account](#).

Enter Your Student ID

Student ID:

[View Profile](#)

Figure 11: Profile page when If the student ID provided does not exist.

END OF QUESTION PAPER